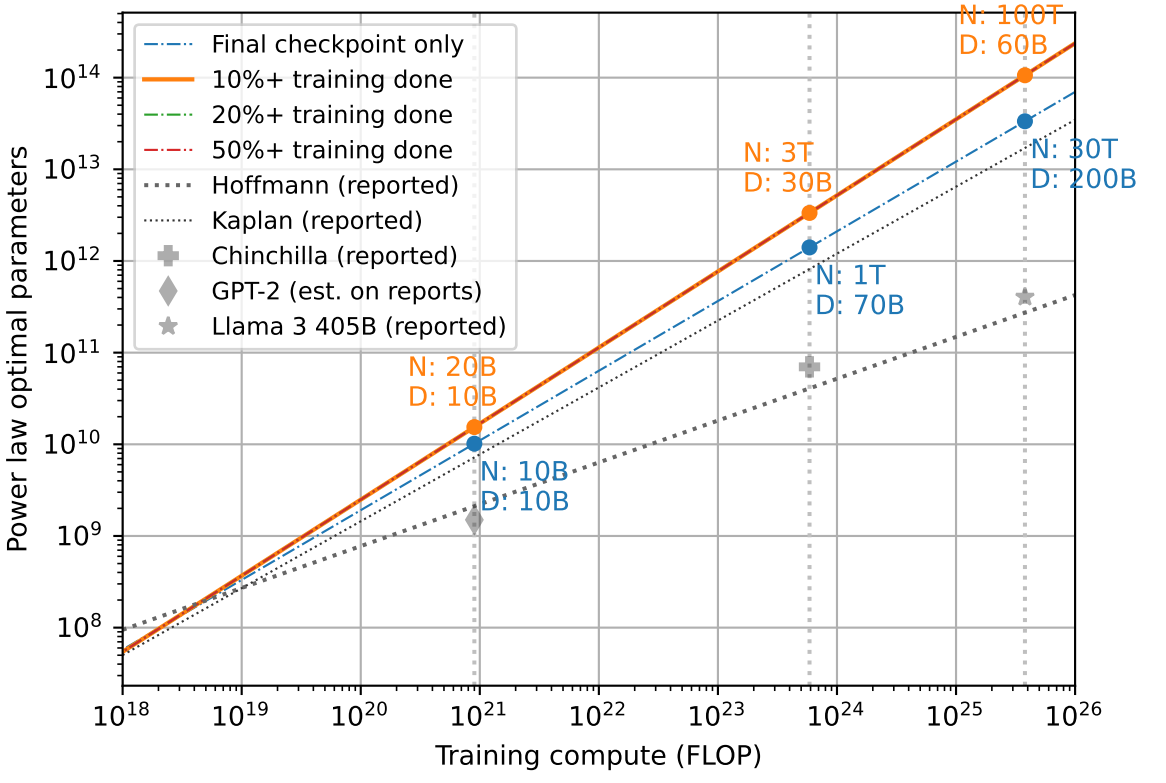
Final checkpoint only
10%+ training done
20%+ training done
50%+ training done
Hoffmann (reported)
Kaplan (reported)
Chinchilla (reported)
GPT-2 (est. on reports)
Llama 3 405B (reported)
N: 20B
D: 10B
N: 10B
D: 10B
N: 3T
D: 30B
N: 1T
D: 70B
N: 100T
D: 60B
N: 30T
D: 200B
Power law optimal parameters
Training compute (FLOP)
10^{8}
10^{9}
10^{10}
10^{11}
10^{12}
10^{13}
10^{14}
10^{18}
10^{19}
10^{20}
10^{21}
10^{22}
10^{23}
10^{24}
10^{25}
10^{26}